

## **Shriners Gait Model Setup (Currently Vicon Use Only)**

### **1. Moving VST files to correct location for Vicon (Vicon Skeletal Models)**

- a. Copy the .vst files from the vsts folder in the GitHub project (ViconClinicalGait/vsts/)
  - i. Current vsts included with the project are:
    1. SMACNet no KAD.vst (Standard Model)
- b. Paste into your Vicon ModelTemplates folder
  - i. Example filepath  
C:\Users\Public\Documents\Vicon\Nexus2.x\ModelTemplates

### **2. Moving Pipeline files to correct location for Vicon**

- a. Copy the .Pipeline files from the Utils folder in the GitHub project (ViconClinicalGait/Utils/Pipeline)
  - i. Current Pipelines included with the project are:
    1. Py3\_Static
    2. Py3\_Dynamic
    3. Py3\_Static\_AFO
- b. Paste into your Vicon Pipelines folder
  - i. Example filepath  
C:\Users\Public\Documents\Vicon\Nexus2.x\Configurations\Pipelines

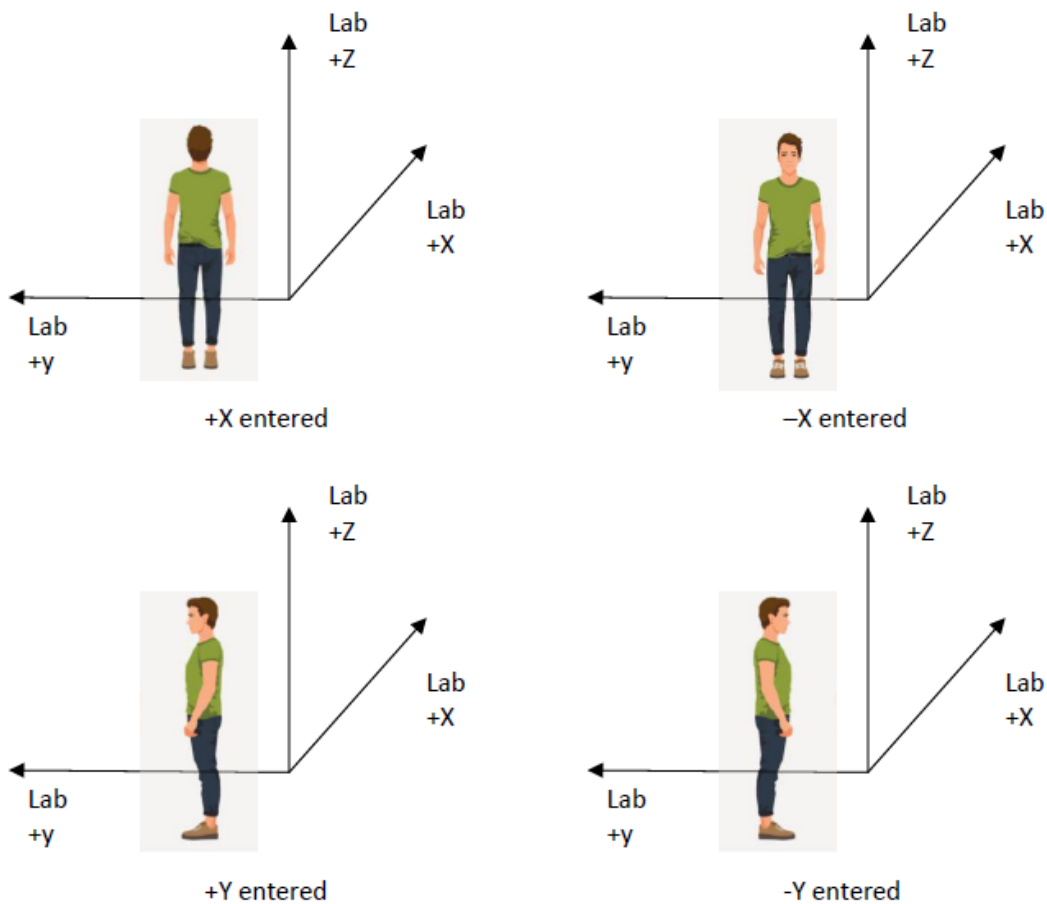
### **3. Moving Python Code to the correct location for Vicon**

- a. Copy the .py files from the Py3\_ShrineGaitModel folder in the GitHub project (ViconClinicalGait/Py3\_ShrineGaitModel)
  - i. Current Python files included with the project are:
    1. Py3\_CreateGCD
    2. Py3\_DynamicMain
    3. Py3\_GaitModules
    4. Py3\_MathModules
    5. Py3\_StaticMain
    6. Py3\_UserPreferences
- b. Paste into a Vicon Python folder
  - i. Example filepath: C:\Users\Public\Documents\Vicon\Python
- c. Possible errors in where these files are placed could impact their use
- d. File location matters significantly, the Py3\_UserPreferences.py file must be in the same directory(folder) as the Py3\_StaticMain.py file for it to identify the appropriate file.

### **4. Updating the User Preferences File:** Several items need to be reviewed in "UserPreferences.py" for each individual lab, these are the contained in the first 7 lines of code

- a. The marker diameter needs to be entered for your lab
  - i. Most likely the diameter will be 9.5 mm

- ii. If you don't use a base that adds thickness the values entered will be:
  - 1. `self.MarkerDiameter = 9.5`
- iii. If you use a base, add the thickness of the base and multiply by 2. For example if you use a 2 mm base add the value of 4 as follows:
  - 1. `self.MarkerDiameter = 9.5+4`
- b. For the next line of code (`self.StaticForwardDirection =`) you will need to know the global axis coordinate system for your lab.
  - i. Determine the global axis coordinate system
  - ii. Determine the direction that participants stand during the static trial (+X, -X, +Y, -Y). Enter the direction the participant stands with respect to the lab coordinate system:
  - iii. Here are 4 examples:



- iv. Enter the correct '+X' or '-X' or '+Y' or '-Y' for your lab and data collection protocol: e.g. `self.StaticForwardDirection = '-X'`
- c. The 3rd line of code is related to the selection of the hip model implementation (`self.HipModelName`)
  - i. The default for clinical gait analysis for Shriners Children's is 'HarringtonHybrid'

- ii. 'HarringtonHybrid' uses regression equations to determine the x,y,z coordinates of the hip joint centers. The full set of equations use Pelvic Width (measured distance between ASIS during physical exam), Pelvic Depth (distance between Midpoint of ASIS markers and Midpoint of PSIS markers), and Leg Length from physical exam. **BOTTOM LINE: leg length and ASIS-width need to be measured on physical exam and used.**
- d. The 4th and 5th lines establish the rotation sequence for the pelvis and trunk. The 2 options are the orders ROT (Rotation-Obliquity-Tilt) or TOR (Tilt-Obliquity-Rotation). **'ROT' is the setting that should be used.**
  - i. self.TrunkRotationSequence = 'ROT'
  - ii. self.PelvisRotationSequence = 'ROT'
- e. The 6th line sets the shank coordinate system as Distal or Proximal. **The setting of 'Distal' should be used.** NOTE: Distal or Proximal must start with capital letter
  - i. self.ShankCoordinateSystem = Distal
- f. Lines 7 to 70 establish the marker labels used in the VST files. The labels in the py3\_UserPreferences.py file match these labels as needed, so this part of the file does not need to be updated unless you decide to change the marker labels.

## 5. Static Trial

- a. In session folder, create subject and attach the appropriate vst (Standard Modell)
  - i. Under Subject Properties, enter all required anthropometrics (highlighted in red)
  - ii. SoleDelta is the difference in vertical distance at the heel of the shoe and the toe of the shoe for shod trials (better representation of what the foot is doing in the shoe)
- b. Record a static trial
  - i. Open static
  - ii. Run "Reconstruct and Label" pipeline. Check marker labels

| General                       |                                |
|-------------------------------|--------------------------------|
| Bodymass (kg):                | <input type="text"/>           |
| Height (mm):                  | <input type="text"/>           |
| InterAsisDistance (mm):       | <input type="text"/>           |
| Left                          |                                |
| LegLength (mm):               | <input type="text"/>           |
| AsisTrocanterDistance (m...): | <input type="text"/>           |
| KneeWidth (mm):               | <input type="text"/>           |
| AnkleWidth (mm):              | <input type="text"/>           |
| TibialTorsion (deg):          | <input type="text" value="0"/> |
| SoleDelta (mm):               | <input type="text" value="0"/> |
| Right                         |                                |
| LegLength (mm):               | <input type="text"/>           |
| AsisTrocanterDistance (m...): | <input type="text"/>           |
| KneeWidth (mm):               | <input type="text"/>           |
| AnkleWidth (mm):              | <input type="text"/>           |
| TibialTorsion (deg):          | <input type="text" value="0"/> |
| SoleDelta (mm):               | <input type="text" value="0"/> |

c. Run “Py\_3 Static” pipeline

i. On the following screen:

1. Fill out required patient/subject information (outlined in red)
2. Match Static Frame Number to the selected frame number of the static trial (outlined in blue). Static frame should be fully labeled.
3. Check that all required anthropometric measures are filled in (outlined in green)
4. Ignore the foot model parameters (a separate user guide is included for using the multisegment foot model)

The screenshot shows the 'Static' software interface with the following sections:

- Patient Information:** Contains fields for First Name, Last Name, Patient Number, Date Of Birth (01 Jan 1981), Trial Activity (Static), Trial Modifier (Barefoot), Static Forward Direction (+X), and Assistive Device (None). These fields are outlined in red.
- Anthropometric Parameters:** Contains fields for Static File, Anthropometric File (Static\_BF\_2025Consistency), Date Of Data Capture (17 Dec 2025), and Static Frame Number (85). The Static Frame Number field is outlined in blue. Below these are anthropometric measures: BodyMass (80.0 kg), Height (1800 mm), ASIS-to-ASIS Distance (248 mm), Leg Length (995 mm Left, 995 mm Right), Knee Width (127 mm Left, 126 mm Right), Ankle Width (78 mm Left, 78 mm Right), and ASIS-to-GT Distance (80 mm Left, 80 mm Right). These fields are outlined in green.
- Foot Model:** Contains checkboxes for Left and Right, and various angle measurements for Hindfoot Varus/Valgus, Calcaneal Pitch, Hindfoot Progression, First Metatarsal Pitch, and Forefoot Progression.
- Special Test Conditions:** Contains checkboxes for Left and Right, and options for Subject Plantigrade during Static Trial, Subject Shod (with or without orthoses), Knee Option (KAD or M/L Markers), Difference in sole thickness between heel and toes (mm), and Apply Pelvic Fix Option.

Buttons on the right side include 'Save & Proceed' and 'Quit'.

5. Under “Special Test Conditions”, select appropriate boxes to indicate whether subject is plantigrade and whether subject is shod. If shod, enter sole delta in mm (change in height of shoe sole between toe box and heel (positive number or 0))

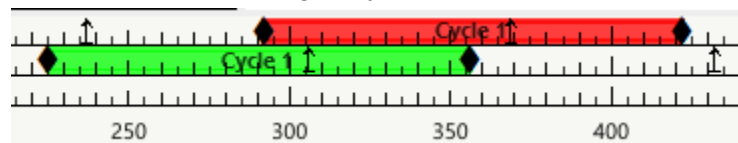
6. We recommend not changing the knee option setting. This model will work with a KAD (knee alignment device), but we have shifted to using medial and lateral knee markers
  7. Ignore the Pelfix option (a separate user guide is included for using this option)
  8. Click “Save & Proceed”
- ii. Compare marker measurements to clinical measurements (The Shriners Gait Model uses clinical measurements to calculate joint centers and virtual markers). Marker measurements should be relatively close to clinical measures in most cases (5-10 mm for knees and ankles, around 20-25 mm for hip width). More significant differences should be evaluated to determine if marker placement is in line with the clinical measurement. Click “Proceed”
  - iii. Check Standing Posture measurements to see if they align with patient position and clinical impression. When using just the Standard Model, the Foot Segment measurements will be empty. Click “Save Results” and “Save PDF”. Click “Quit”
    1. “Save Results” saves a static.py file that contains all of the static trial information needed to process the dynamic trials (marker relationships between technical and anatomical coordinate systems, anthropometrics, etc.)
    2. “Save PDF” saves a pdf output report of the static trial you just processed, showing the marker measurement comparison, standing posture measurements, etc.

## 6. Dynamic Trials

- a. Medial knee and medial malleoli markers can be removed for dynamic trials
- b. Record dynamic walking trials in your capture volume

## 7. Processing Dynamic Trials

- a. Open dynamic walking trial
- b. Run “Reconstruct & Label” pipeline
  - i. Check marker labels and gap fill as needed
  - ii. Crop trial as needed. The Shriners Gait Model requires all markers to be present on the first frame of the trial
- c. Add in gait events
  - i. Label at least one full gait cycle as shown below



- d. This model does support kinetics requiring force plates to be integrated with Vicon
  - i. Assign each force plate based on which foot strikes the plate (right or left)

- e. Run "Py3\_Dynamic" pipeline

## **8. Outputs**

- a. The Py3\_Dynamic pipeline will complete the calculations and save the results (kinematics, kinetics, etc.) as model outputs
- b. These outputs can be found in Vicon by:
  - i. clicking on the Subjects Tab
  - ii. clicking the drop down menu to the left of the subject name
  - iii. clicking the drop down menu to the left of Model Outputs
  - iv. and clicking the drop down menu to the left of the model output you are interested in
- c. The pipeline will also output a .GCD file
  - i. This is a type of text file that contains each model output time normalized to be represented as percentage of the gait cycle (101 points 0-100%)
  - ii. This text file can be used as a coding input to visualize the data
  - iii. No visualization code has been provided